

the population of states (POP). When a model that relates the dependent variable EXP to the independent variables AID, INC and POP was estimated by ordinary least squares, using census data for 50 states, the following results were obtained (with t statistics in parentheses):

$$\begin{aligned} \text{EXP} = & -46.81 + 0.00324 \text{ AID} + 0.00019 \text{ INC} - 0.597 \text{ POP} \\ & (-0.56) \quad (13.64) \quad (8.12) \quad (-5.71) \\ R^2 = & 0.993 \quad F = 2190 \end{aligned}$$

There is an important possible source of measurement error in the AID variable. This component can be subject to substantial error as the state aid programmes involves fixed sums of money, and the money involved are easy to measure even before state and local budgets are set.

We can test the presence of measurement error by using a Hausman specification test. To perform the test, we use the population of primary and secondary school children (PS) as an instrument. The test proceeds in two stages. In the first stage AID is regressed on PS, and the residual variable $\hat{w}_i$ is calculated as follows:

$$\begin{aligned} \hat{w}_i = & \text{AID} - 77.95 + 0.845 \text{ PS} \quad R^2 = 0.87 \\ & (-1.28) \quad (18.02) \end{aligned}$$

In the second stage $\hat{w}_i$ is added to the original regression to correct for measurement error. The resulting equation is

$$\begin{aligned} \text{EXP} = & -138.51 + 0.00174 \text{ AID} + 0.00018 \text{ INC} - 0.275 \text{ POP} + 1.372 \hat{w}_i \\ & (-1.41) \quad (1.94) \quad (7.55) \quad (-1.29) \quad (1.73) \end{aligned}$$

A two-tailed t test of the null hypothesis that there is no measurement error would be accepted at the 5 per cent level, since $1.73 < 1.96$. However, measurement error would be important if we were using either a one-tail test, or a two-tailed test at 10 per cent significance level. Note that correcting for the possibility of measurement error has substantially lowered the coefficient on the AID variable, suggesting that measurement error caused the effect of AID on spending to be overstated.

12.5 INVERSE REGRESSION

When we have the variables y and x both measured with errors (the observed values being y^* and x^*), we consider two regression equations:

- 1) Regression of y^* on x^* which is called the “direct” regression.
- 2) Regression of x^* on y^* which is called the “inverse” and “reverse” regression.

Inverse regression has been frequently advocated in the case of analysis of salary discrimination. Since the problem is one of usual errors in variables, both regressions need to be computed and whether inverse regression alone gives the correct estimates depends on the assumptions we make.

The usual model, in its simplest form, is that of two explanatory variables, one of which is measured with error

Violations of Basic Assumptions

$$y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

where y = salary

x_1 = true qualifications

x_2 = gender

What we are interested in is the coefficient of x_2 . The problem is that x_1 is measured with error.

Let x_i^* = measured qualifications

$$x_i^* = x_i + v_i$$

Suppose we adopt the notion that

$$x_2 = \begin{cases} 1, & \text{for men} \\ 0, & \text{for women} \end{cases}$$

Then $\beta_2 > 0$ implies that men are paid more than women with the same qualification and thus there is gender discrimination. A direct least squares estimation of eq. (12.22) with x_1^* substituted for x_1 and $\hat{\beta}_2 > 0$ has been frequently used as evidence of gender discrimination.

In the inverse regression we take

$$x_1^* = \gamma_1 y + \gamma_2 x_2 + w \quad \dots (12.23)$$

We are asking whether men more or less qualified than women having the same salaries. The proponents of inverse regression argue that to establish discrimination, one has to have $\hat{\gamma}_2 < 0$ in eq. (12.23); that is, among men and women receiving equal salaries, men possess lower qualifications.

The usual errors in variables results however show that we should not make inferences on the basis of $\hat{\beta}_2$ and $\hat{\gamma}_2$ but obtain bounds for β_2 from the direct regression and the inverse regression estimates.

Check Your Progress 2

- 1) Show that in the two-variable model $\hat{\beta} = \frac{\sum y_i z_i}{\sum z_i^2}$ (where z is an instrument) will not yield a consistent estimate of the true slope parameter.

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- 2) What is the Hausman specification test? How would you carry out a Hausman specification test to evaluate the presence or absence of measurement error?

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12.0 LET US SUM UP

In ordinary least squares model, we assume that sample observation are measured without error, which is always not true. When this assumption does not hold, OLS estimators are biased and inconsistent. Errors may appear in the measurement of dependent variable, independent variable or both. When there is error in dependent variable, this does not destroy the unbiased property of the OLS estimators, but the estimated variances are larger than the case where there is no such errors of measurement. On the other hand, when there is error in the independent variable, or both dependent and independent variables, then OLS estimates are not only biased but also inconsistent.

We can test for the presence of measurement error by using the Hausman specification test. One technique which is available and can solve the measurement error problem is the technique of instrumental variables estimation.

12.7 KEY WORDS

- Asymptotic Bias** : When the classical least-squares estimator is biased even with the increase in sample size, it is known as asymptotic bias or inconsistency.
- Equation Error** : It represents the influence of various explanatory variables that have not accurately been included in the model and is in the form of disturbance term.
- Errors in Variables** : When errors are found in the measurement of variables, we say that there are errors in variables. These errors can substantially alter the properties of the estimated regression parameters.
- Explanatory Variable** : It is the independent variable which explains the dependent variable.
- Instrumental Variable** : It is a random variable, which is highly correlated with the independent variable and at the same time uncorrelated with the error term in the equation.
- Inverse Regression** : It is the reverse of direct regression. If we consider regression of Y on X as direct regression, then regression of X on Y is inverse regression.
- Measurement Error** : When either dependent variable, or independent variable or both, are measured subject to error, we say that there is measurement error.
- Parameter** : It is a measure of some characteristics of the population.

Proxy Variable	: It is a variable, which is proxy for the true variable.
Specification Error	: Errors found in the specification of the model.
Stochastic Disturbance Term	: It is the random error term whose value is based on an underlying probability distribution.

12.8 SOME USEFUL BOOKS

Intriligator, Michael, Ronald Bodkin and Cheng Hsiao (1996); *Econometric Models, Techniques, and Applications*, Prentice-Hall International Inc., New Jersey.

Johnston, J., Dinardo, J., 2007, “*Econometric Methods*”, 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125

Maddala, G. S. and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley

Pindyck, Robert S. and Daneil L. Rubinfeld (1998); *Econometric Models and Economic Forecasts*, Fourth Edition, Irwin/McGraw-Hill, Singapore.

Verbeek, M. 2012, “*A Guide to Modern Econometrics*”, Fourth Edition, John Wiley & Sons

12.9 ANSWERS/HINTS OF CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Go through Sections 12.1 and 12.3.
- 2) a) true b) true.
- 3) Go through Section 12.3.

Check Your Progress 1

- 1) Go through Section 12.4.
- 2) Go through Section 12.5 and Example 12.2.

UNIT 13 STOCHASTIC REGRESSORS*

Structure

- 13.0 Objective
- 13.1 Introduction
- 13.2 Endogeneity Problem
 - 13.2.1 Omitted Variables
 - 13.2.2 Measurement Error
 - 13.2.3 Simultaneity
- 13.3 Instrumental Variable Estimator
- 13.4 Two-Stage Least Squares Estimator
- 13.5 Let Us Sum Up
- 13.6 Key Words
- 13.7 Some Useful Books
- 13.8 Answers/ Hints to Check Your Progress Exercises

13.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the endogeneity problem due to omitted variables, measurement error, and simultaneity;
- explain the need for instrumental variables and how they can be used; and
- describe the two-stage least squares estimator.

13.1 INTRODUCTION

While discussing the classical regression model (see Unit 4), we assumed that the values of the explanatory variable, x_i , are kept fixed in repeated samples. An implication of the above is that the values assumed by x_i are not random. Consequently, there is no relationship between the stochastic error term u_i and the explanatory variable x_i . This assumption is concisely written as $E(x_i u_i) = 0$.

In real life situations x_i is a random variable, that is, x_i is a stochastic regressor. In such cases $E(x_i u_i) \neq 0$, that is, x_i and u_i are correlated. In this Unit we discuss issues related to the consequences of stochastic regressors and the remedial measures that we should undertake.

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13.2 ENDOGENEITY PROBLEM

Let us consider the following population regression model, which is linear in its parameters

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K + u \quad \dots(13.1)$$

where y : dependent variable

$x_1, x_2, \dots, x_K$: explanatory variables

u : unobservable random disturbance or error

$\beta_1, \beta_2, \dots, \beta_K$: the parameters to be estimated

The key condition on which equation (13.1) under OLS can consistently estimate the β_j is that mean value of the stochastic error is zero and is uncorrelated with each of explanatory variables. Symbolically, we can write:

$$E(u) = 0, \text{Cov}(x_j, u) = 0, \quad j = 1, 2, \dots, K \quad \dots (13.2)$$

In regression analysis, assumption (13.2) is sufficiently supported by the zero conditional mean assumption, which can be written as:

$$E(u | x_1, x_2, \dots, x_K) = E(u | x) = 0 \quad \dots (13.3)$$

The population regression function under equation (13.1) and assumption (13.3) can be written as:

$$E(y | x_1, x_2, \dots, x_K) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K \quad \dots (13.4)$$

An explanatory variable x_j is said to be 'endogenous' in equation (13.1), if it is correlated with the stochastic error term u . Similarly, x_j is said to be exogenous in equation (13.1), if it is not correlated with the stochastic error u . Assumption (13.3) implies that each explanatory variable is exogenous.

An explanatory variable x_j is said to be endogenous if it is correlated with the stochastic error u . Thus, endogeneity problem arises whenever the explanatory variable is correlated with the error term. There are three ways of observing endogeneity in regression models. These are:

- a. Omitted Variables
- b. Measurement Error
- c. Simultaneity

13.2.1 Omitted Variables

The case of omitted variables was discussed in Unit 8. It arises when we are unable to include certain additional variables in our regression model due to data unavailability. Let us suppose that $E(y|x, q)$ is the conditional expectation function which is linear in parameters and additive in q (suppose, q represents the excluded variables). We always estimate $E(y|x)$ for unobserved q . But such a regression

model is different from $E(y|x, q)$, even if q and x are correlated. In the case of excluded variables, we are writing equation (13.1) and q gets included in the stochastic error u . Thus, variable x_j is endogenous in a regression model if q (i.e., excluded variables) and x_j are correlated.

13.2.2 Measurement Error

In regression analysis we implicitly assume that the dependent and independent variables are measured without any error. That is, the variables are exact, and error is zero. However, this is only an assumption and in real life errors creep in due to various reasons such as non-response error, reporting error, and computing error. We discussed the issue of error-in-variables in Unit 12.

Whatever be the reason, measurement error in variables lead to biased and inefficient estimators. The observed variables of the sample regression function on which we collect data are modelled as:

$$\tilde{x} = x + u \quad \dots (13.5)$$

$$\tilde{y} = y + v \quad \dots (13.6)$$

Equation (13.5) and equation (13.6) show that our observed variables $\tilde{x}$ and $\tilde{y}$ are measured with additive errors u and v respectively. The measurement error in x creates an endogeneity bias because the measurement error becomes a part of the error term u in the regression equation. This can be viewed from equation (13.5) as follows:

$$x = \tilde{x} - u$$

Now putting the value of x in $y = \beta x + \epsilon$ we can write:

$$y = \beta(\tilde{x} - u) + \epsilon$$

$$\text{or, } y_i = \beta\tilde{x}_i + (\epsilon - \beta u) \quad \dots (13.7)$$

Because of the positive correlation between $\tilde{x}$ and u observed in equation (13.5), the OLS estimation produces certain bias in $\tilde{\beta}$. Such bias may be negative or positive depending upon the sign of the true β in equation (13.7). If the true β is positive, then the bias is negative. On the other hand, if the true β is negative, then bias is positive.

13.2.3 Simultaneity

If any of the explanatory variables is determined simultaneously along with y , then the problem of simultaneity arises. This problem is also called reverse causality, that is, the explanatory variable (x) is caused by the explained variable (y). For example, x_K and u are correlated if x_K is determined partly as a function of y .

Let us consider a two variable regression model ($y_i = \beta x_i + u_i$), where the dependent variable is 'earnings (y_i)' of a person and the independent variable is 'years of schooling (x_i)'. If we analyse the model, there are several variables such as ability of the person, type of school one attended, parental education,

peer-group of the person, etc. that are excluded from the model but influence earnings. All these excluded variables are clubbed together in the stochastic error term. According to this model, if there is no association between x and u , a unit change in x will result in βx change in y . Let us consider the variable 'ability' (which cannot be measured) that is clubbed with u . The variable 'ability' influences y positively – a person with higher ability will have higher value of y . But we know that a person with high ability will have higher value of 'years of schooling', i.e., x and u are correlated. Thus, a person with high ability will influence y in two ways: directly through a higher value of u , and indirectly it will lead to a higher value of x .

Very often it is not possible to distinguish among the three possible forms of endogeneity. In fact, more than one source of endogeneity is observed in any given equation. There are two methods to resolve the endogeneity problem: (i) instrumental variables (IV) method, and (ii) two-stage least squares (2SLS) method. The IV estimator and the 2SLS estimator are consistent, i.e., they approach the value of the parameter as the sample size increases infinitely.

Check Your Progress 1

- 1) What do you mean by endogeneity problem in regression analysis?

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- 2) What are the major sources of endogeneity problems?

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- 3) Explain the problems of measurement error.

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- 4) Explain the problems of omitted variable as an endogeneity problem.

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